Gaurav Agarwal

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Summary

Software engineer with 8+ years of experience across the full development lifecycle, delivering scalable, high-reliability systems in aerospace R&D, startups, and enterprise environments. Driven to build robust, technology-agnostic solutions and contribute to forward-thinking teams solving complex engineering problems.

Work Experience

Boeing India Pvt. Ltd. — *Bengaluru, India*
Software Engineer 3 | Embedded & Avionics Systems — *Nov 2019 – Present*

- Delivered **platform software** across **Cockpit Displays, Common Core, Compute Platform, and Cabin Experience systems** on heterogeneous hardware, building **Linux** and **RTOS**-based real-time services with strong focus on reliability and cross-system integration.
- Designed **scalable backend architectures**, service interfaces, and data models, translating complex system requirements into **modular APIs** and production-ready services while driving standards and design consistency.
- Built **containerized services** and automation workflows on Linux platforms, leading **end-to-end development** from feasibility and architecture through implementation and code reviews.
- Developed **middleware, communication frameworks**, and diagnostic tooling, optimizing performance, fault tolerance, and data handling across **distributed environments** and hardware-in-loop test systems.
- Implemented **CI/CD pipelines** using **GitLab CI** and **Docker**, automating build, integration, and deployment workflows to improve release velocity and system stability across multiple product lines.
- Configured custom hardware and optimized **Board Support Packages**: **U-Boot, kernel, Yocto** recipe-based root file systems, cross-compilation toolchains, and device trees for **secure boot** and runtime performance.

Team Indus (Axiom Research Labs Pvt. Ltd.) — *Bengaluru, India*
Flight Software Engineer | Integrated Avionics | Command & Data Handling — *Jan 2017 – Oct 2019*

- Developed software for **orbital, descent and surface** phases of the **soft landing lunar mission**: onboard state estimation, autonomous attitude correction, terrain tracking, thermal/power control, and **FDIR**.
- Built **Processor in Loop Simulation (PiLS)** framework emulating sensor and actuator electrical interfaces to the Integrated Avionics Unit.

Technical Skills

Languages	C, C++, Python, Java, Bash, SQL, MATLAB, LaTeX
Systems	Linux (Ubuntu, CentOS, Yocto, Buildroot), RTOS (DEOS, VxWorks, FreeRTOS)
Cloud	AWS (EC2, S3, Lambda), Docker, Kubernetes, Nginx, Kafka
DevOps	Git, GitLab CI/CD, Jenkins, CMake, Doxygen, Polyspace, Postman
Protocols	UART, SPI, I2C, CAN, ARINC, TCP/IP, REST, MQTT, gRPC
Backend	PostgreSQL, MongoDB, GraphDB, React.js, OAuth, JWT, Terraform

Education

P.E.S. Institute of Technology (Autonomous, VTU) — *Bengaluru, India*
B.E. in Electrical and Electronics Engineering — *Aug 2013 – May 2017*

- GPA: 8.93 / 10.00

Kerala Samajam Model School — *Jamshedpur, India*
I.C.S.E. & I.S.C. – Pure Science with Computer Application — *Mar 1999 – May 2013*

- ICSE: 93.4%, ISC: 88.75%